



Machining Processes: Drilling

Groover M.P.: Chapters 18 and 19



The student knows:

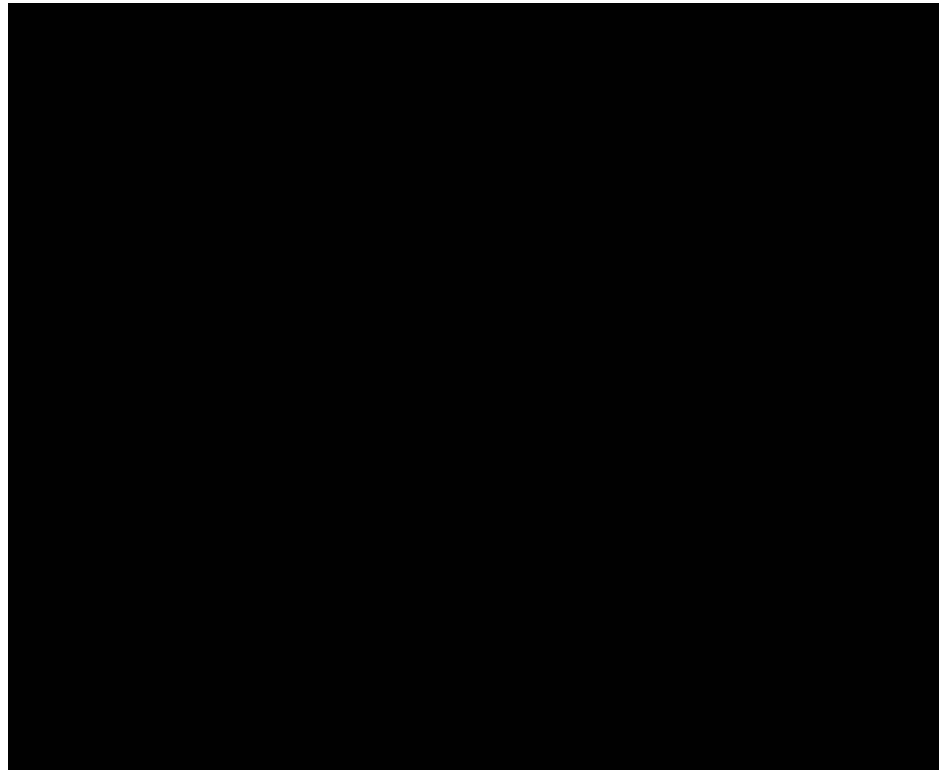
- Basic principles, tools and machine tools;
- How to evaluate cutting force, power and work;
- How to define the appropriate process parameters.





PROCESS OVERVIEW

3

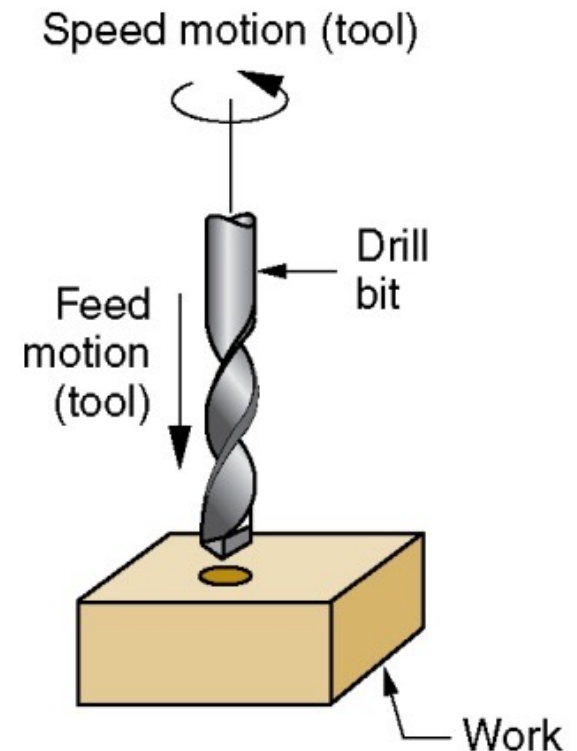
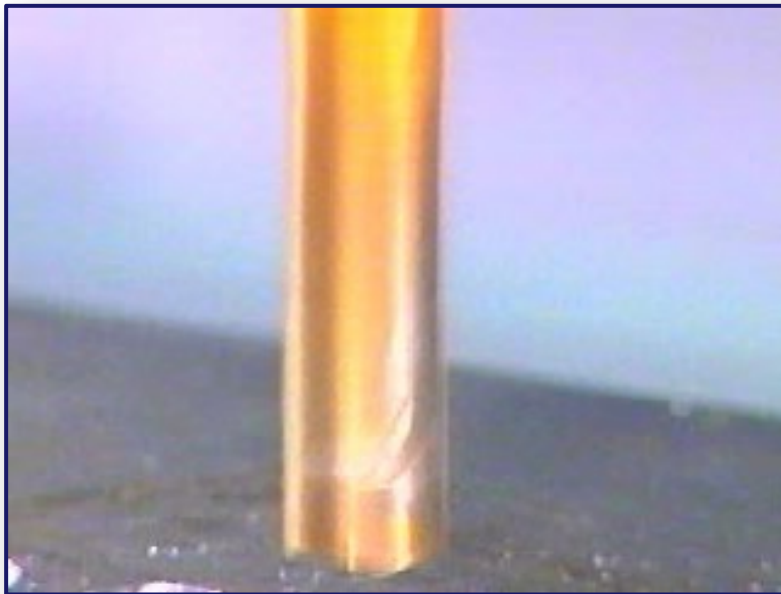




Creates a round hole in a workpiece

Cutting tool called a *drill* or *drill bit*

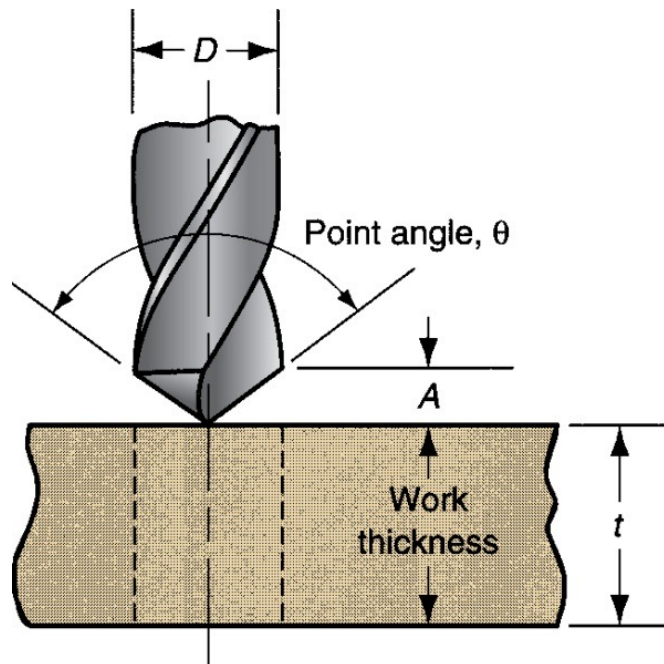
Machine tool: *drill press*



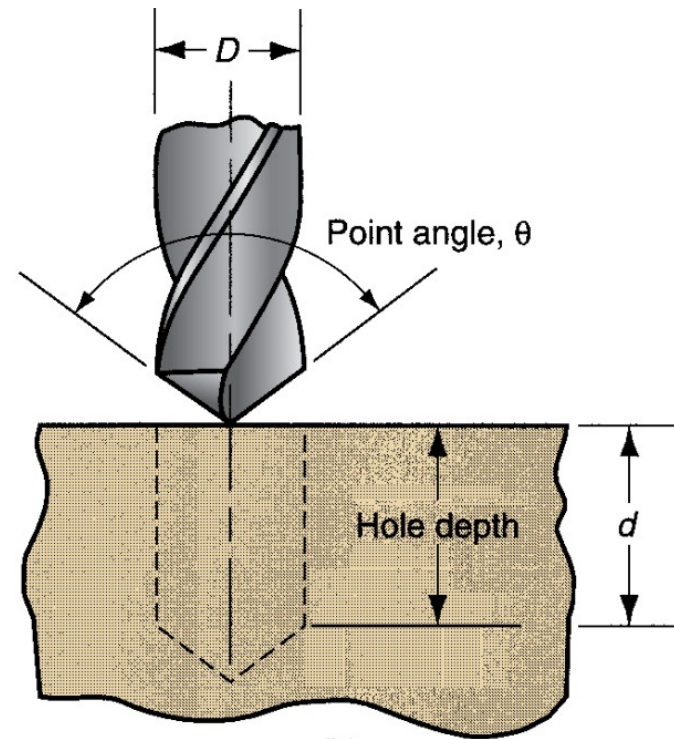


THROUGH HOLE VS. BLIND HOLE

- (a) Through hole - drill exits opposite side of workpiece
- (b) Blind hole – drill does not exit opposite side



(a)



(b)



HOLE TYPES

6

A) Through

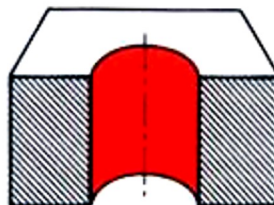
B) Blind

C) Countersink

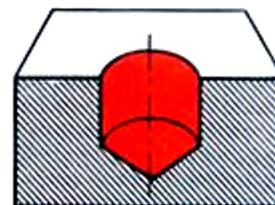
D) Counterbore

E) Conical

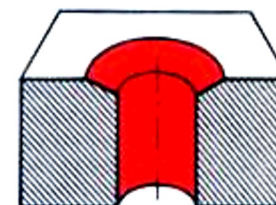
F) Multiple feature



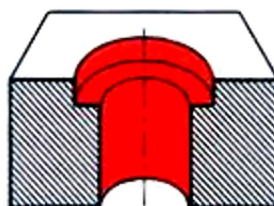
A



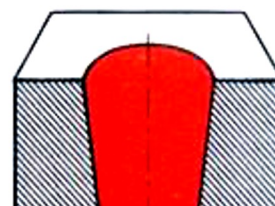
B



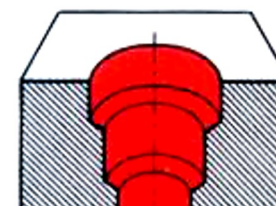
C



D



E



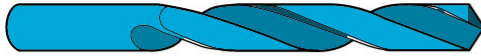
F



OTHER TYPES OF DRILLS

7

(a) Twist drill



(c) Straight-flute drill



(b) Step drill



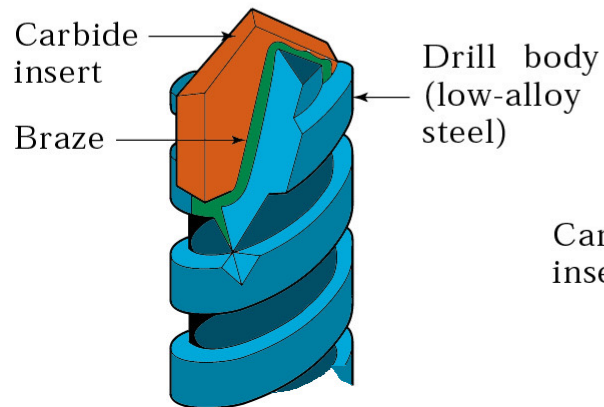
(d) Spade drill



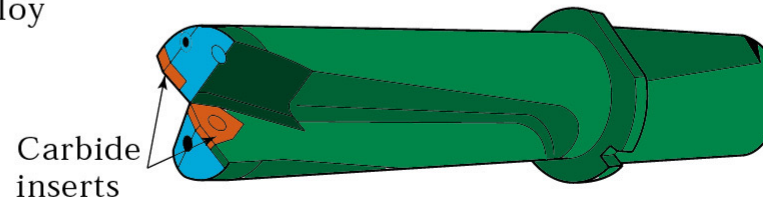
(e) Gun drill



(f) Drill with brazed carbide tip



(g) Drill with indexable carbide inserts





TOOL: TWIST DRILL

8

The helicoidal flutes allow the chip removal.

If the drill has a big diameter, it can be hollow to allow the passage of the cutting fluid to the cutting zone.

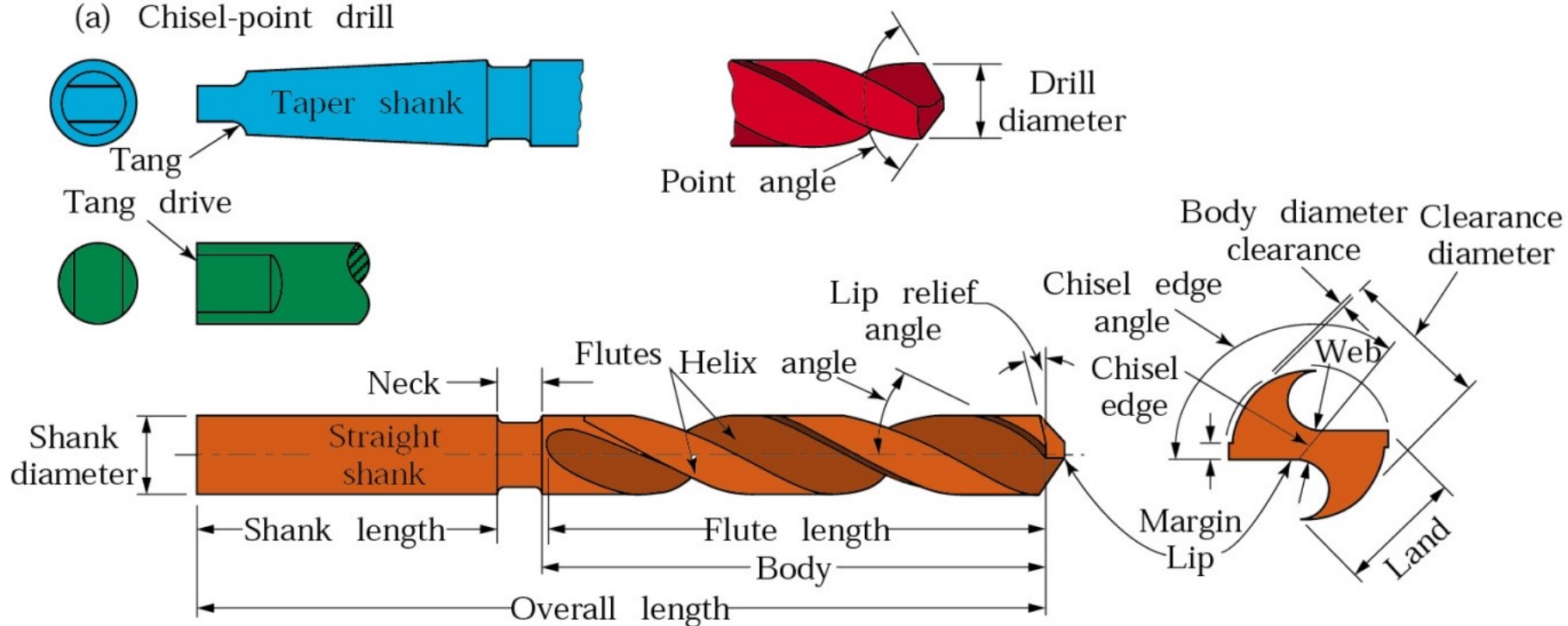


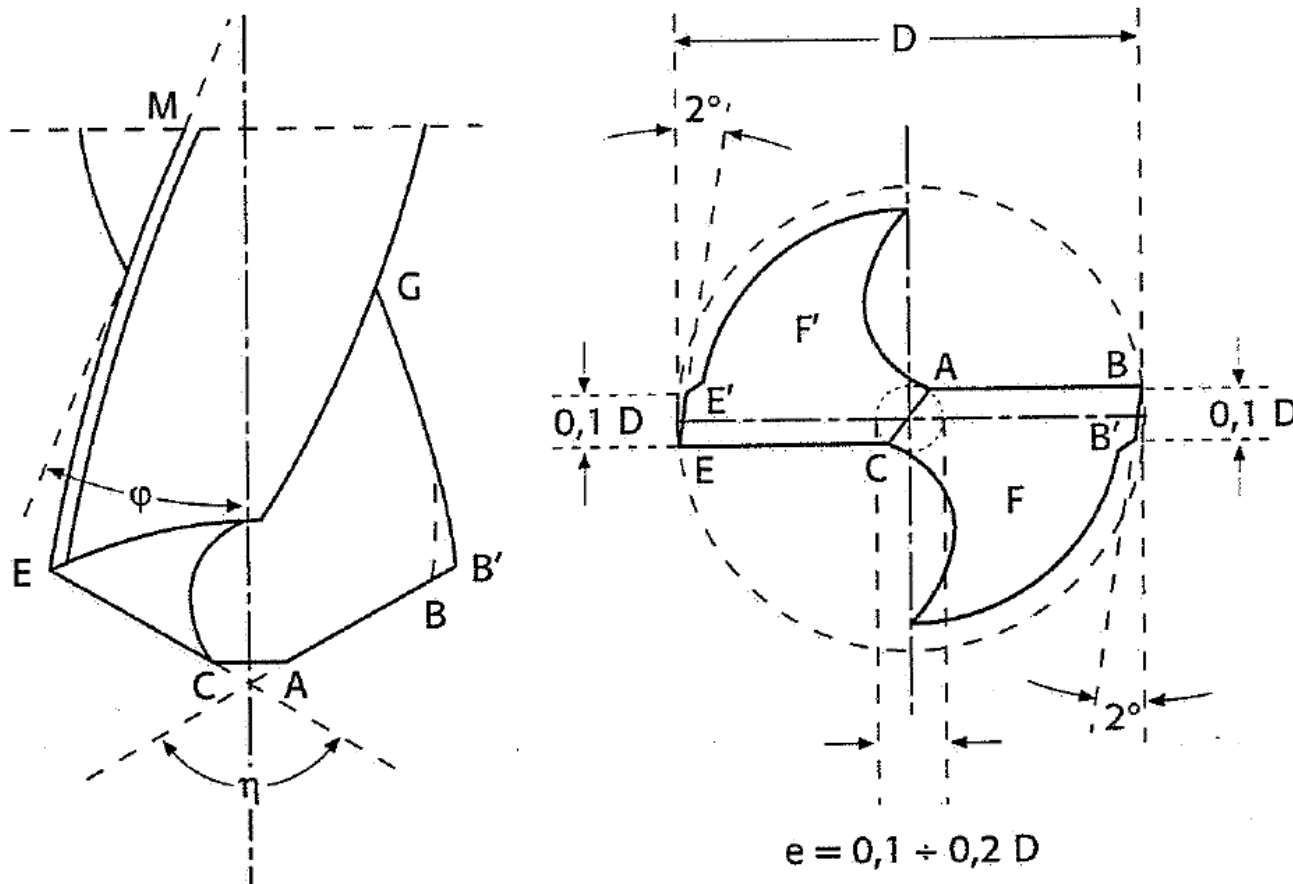


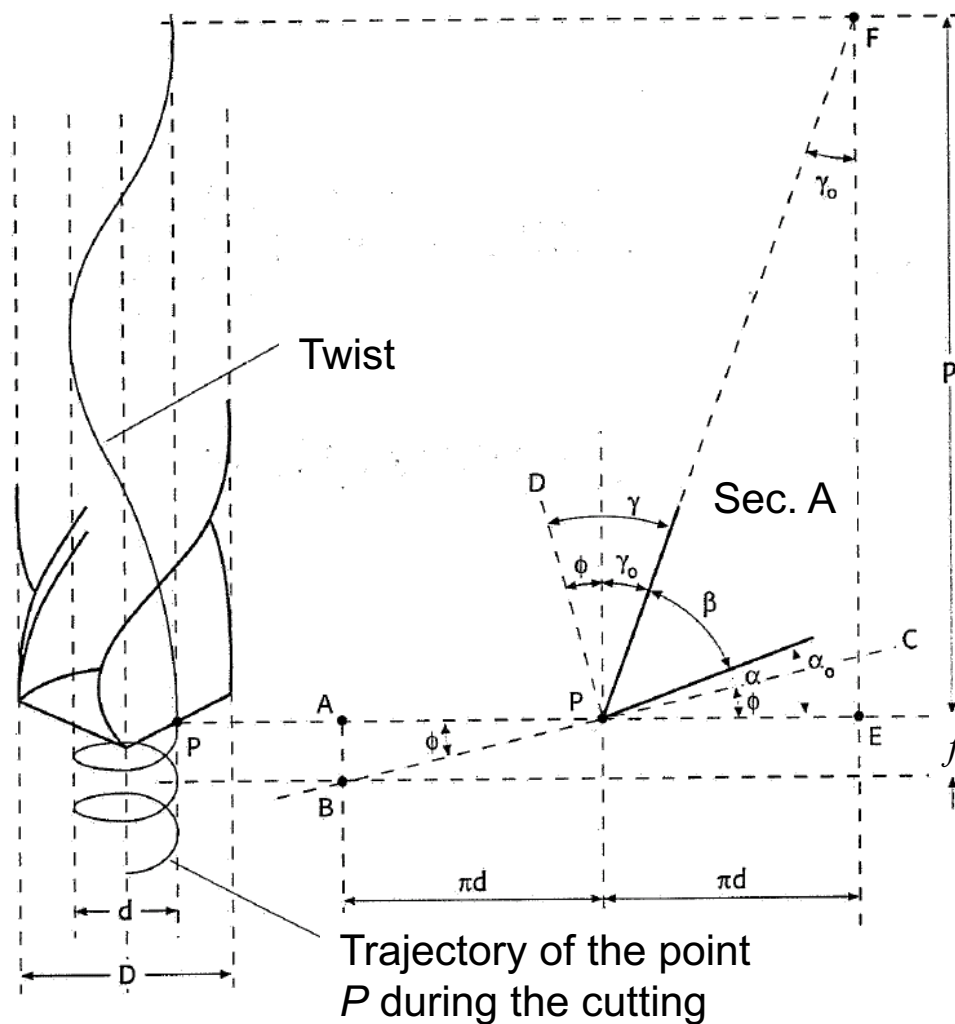
TOOL: TWIST DRILL

9

(a) Chisel-point drill





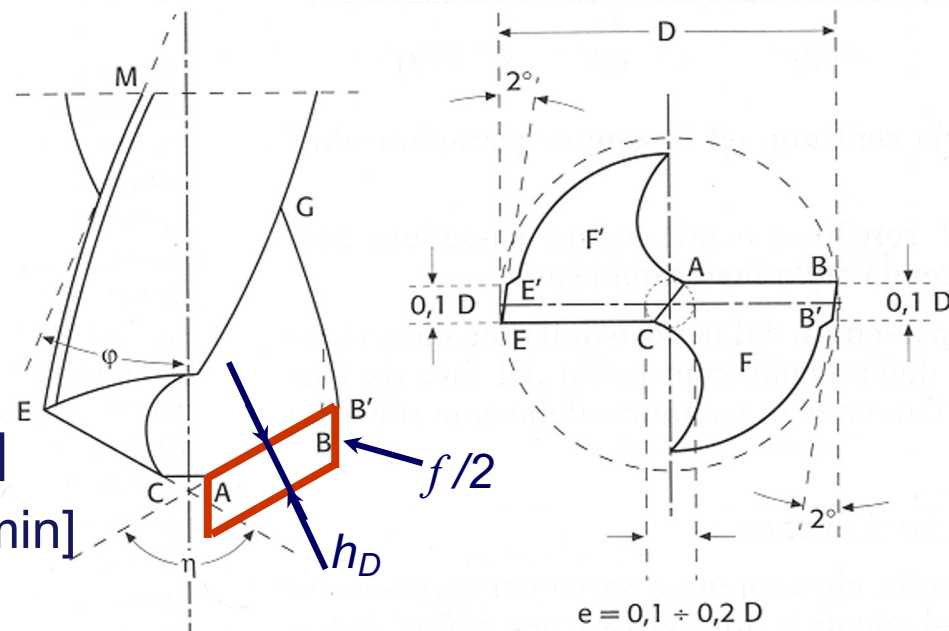


$$\operatorname{tg} \phi = \frac{f}{\pi \cdot d}$$



Characteristic cutting parameters:

- Feed per revolution f [mm/rev]
- Drill diameter D [mm]
- Chip thickness h_D [mm]
- Chip section $A_D = fD/4$ [mm²]
- Rotational speed of the drill n [rpm]
- Cutting speed $v_c = \pi n D/1000$ [m/min]

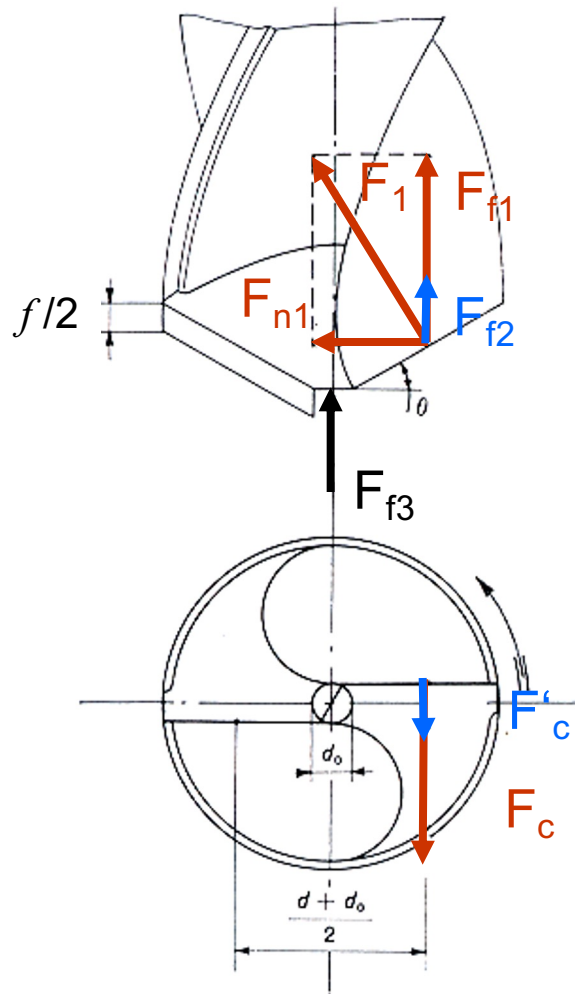


Range:

$$f = 0,01 \div 0,8 \text{ mm/rev}$$

$$D = 1 \div 50 \text{ mm}$$

$$v_c = 5 \div 300 \text{ m/min}$$



1 Forces due to chip formation

2 Forces due to friction on flank faces

3 Force on chisel edge due to feed



Even in drilling we can use the specific pressure method.

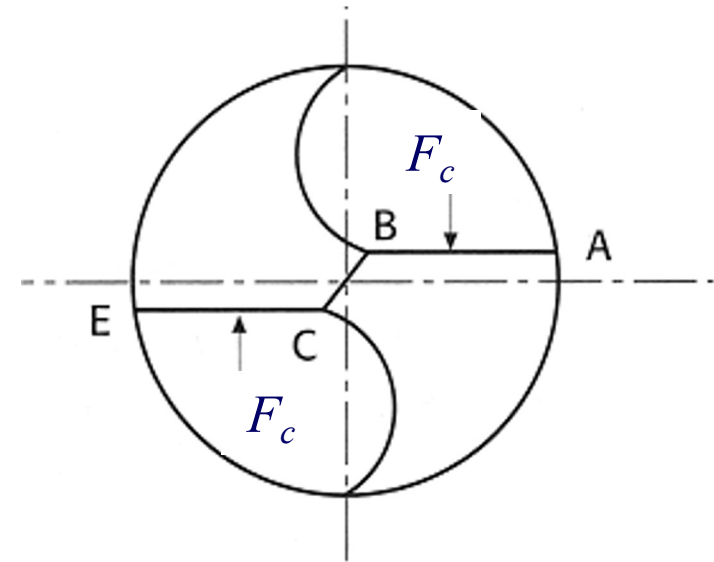
Considered the chip section A_D ,

the F_c force is given by:

$$F_c = k_c A_D \text{ [N]}$$

and then the cutting torque is given by

$$T_c = \frac{F_c \cdot D/2}{1000} = \frac{k_c A_D \cdot D}{2 \cdot 1000} \text{ [Nm]}$$





The cutting power in drilling is given by:

$$P_c = T_c \cdot \omega \quad [\text{W}]$$

since:

$$\omega = \frac{2 \cdot \pi \cdot n}{60}$$

it follows that:

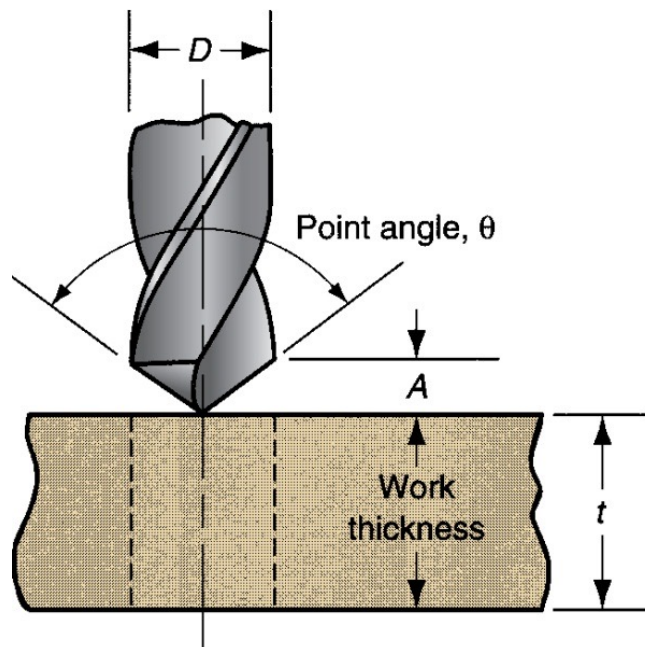
$$P_c = \frac{T_c \cdot \omega}{1000} = \frac{k_c A_D D \cdot \pi \cdot n}{60 \cdot 10^6} \quad [\text{kW}]$$



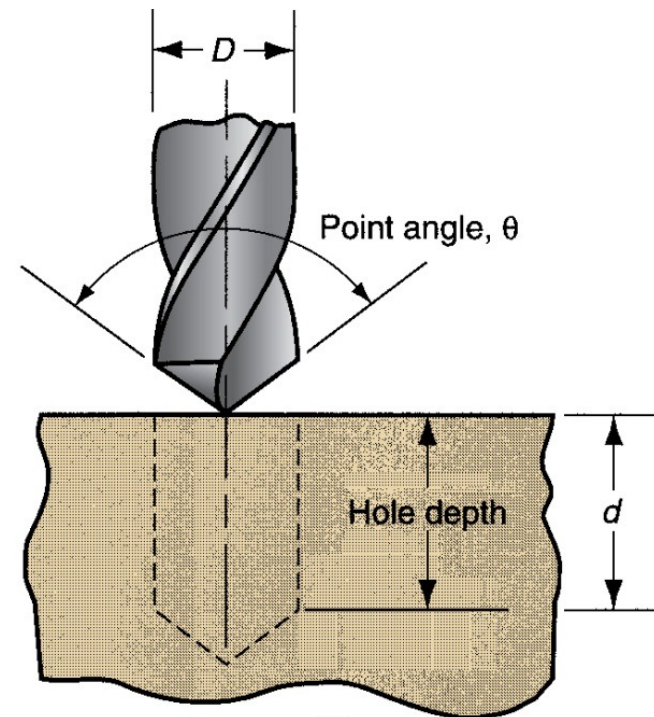
Cutting time:

$$t_c = \frac{L + A}{v_f} = \frac{L + A}{Z f_z n}$$

Where L is the workpiece thickness t (a - through hole) or the hole depth d (b - blind hole).



(a)



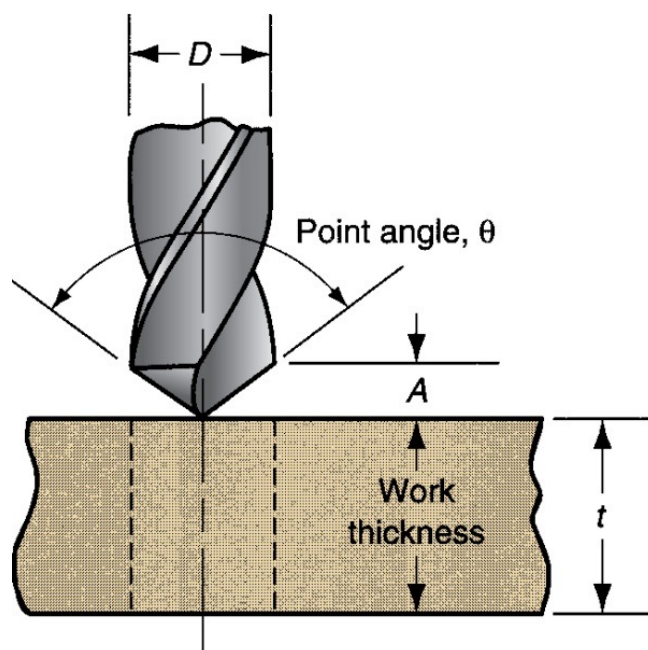
(b)



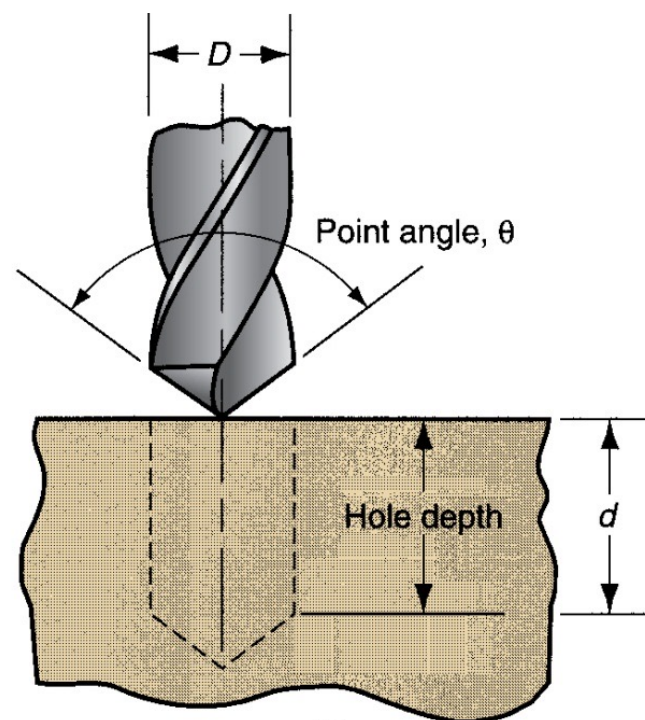
Cutting time:

$$t_c = \frac{V}{MRR}$$

$$MRR = \frac{\pi}{4} D^2 v_f$$



(a)



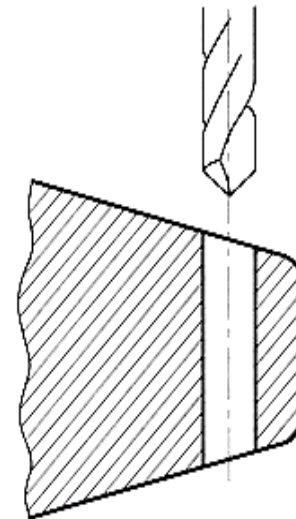
(b)



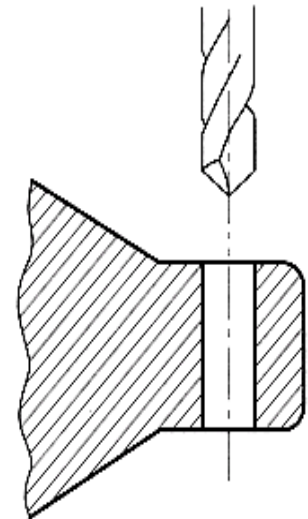
Often a critical problem in drilling is the tight tolerances of the hole.

Some basic rules:

- generate a flat entering surface, normal to the drill axis, to avoid bending of the drill;
- drill the hole in many passes, with a progressive increase of the drill hole.



Poor

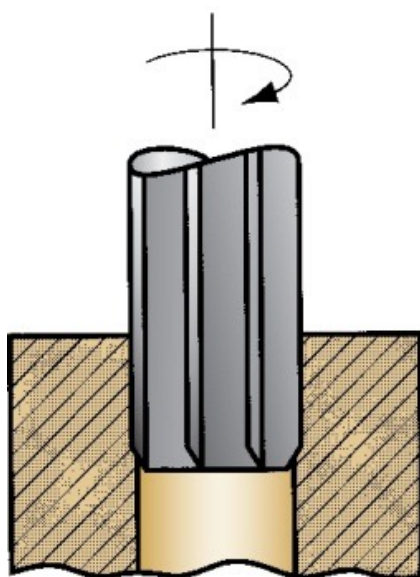


Good

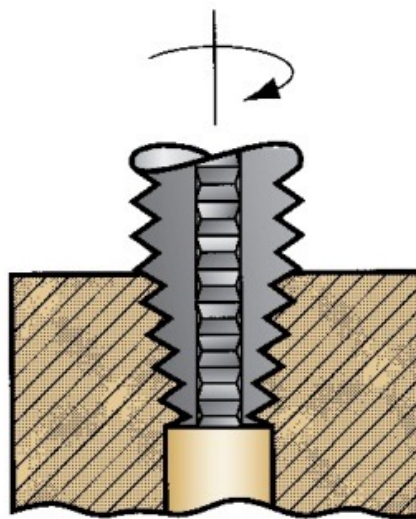


OPERATIONS RELATED TO DRILLING

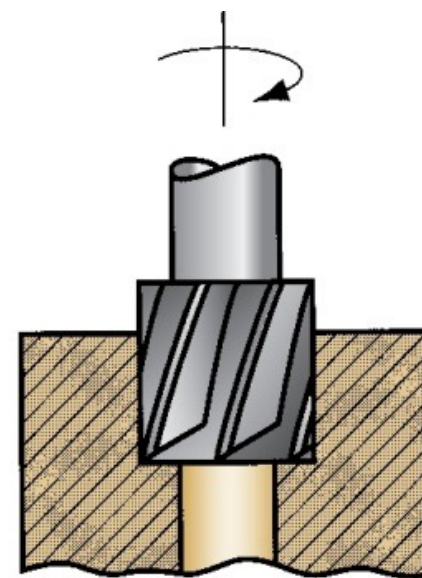
(a) Reaming, (b) Tapping, (c) Counterboring



(a)



(b)

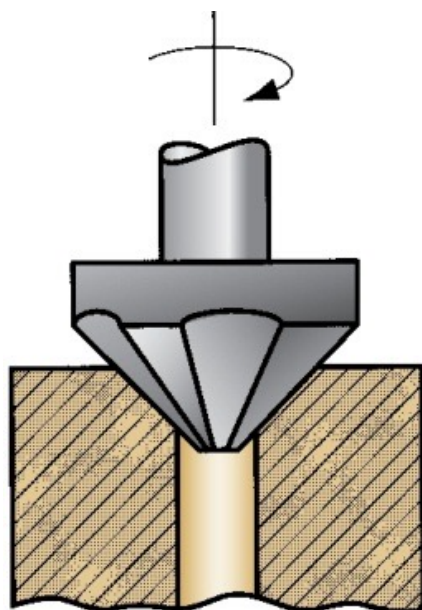


(c)

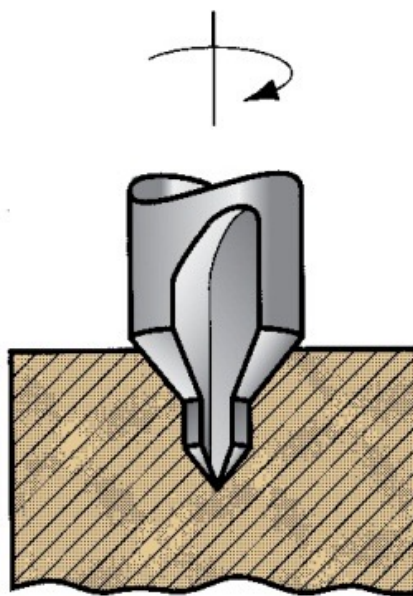


MORE OPERATIONS RELATED TO DRILLING

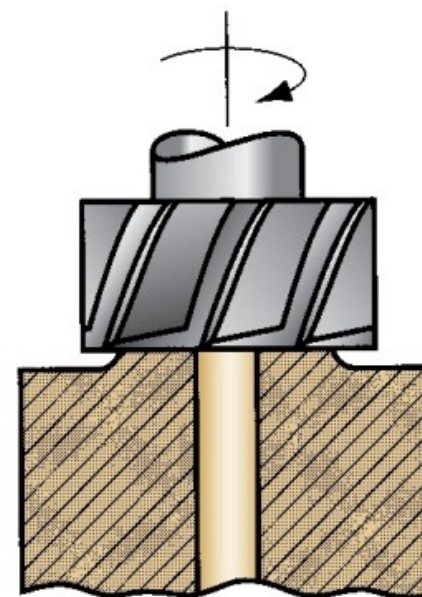
(d) Countersinking, (e) Center drilling, (f) Spot facing



(d)



(e)

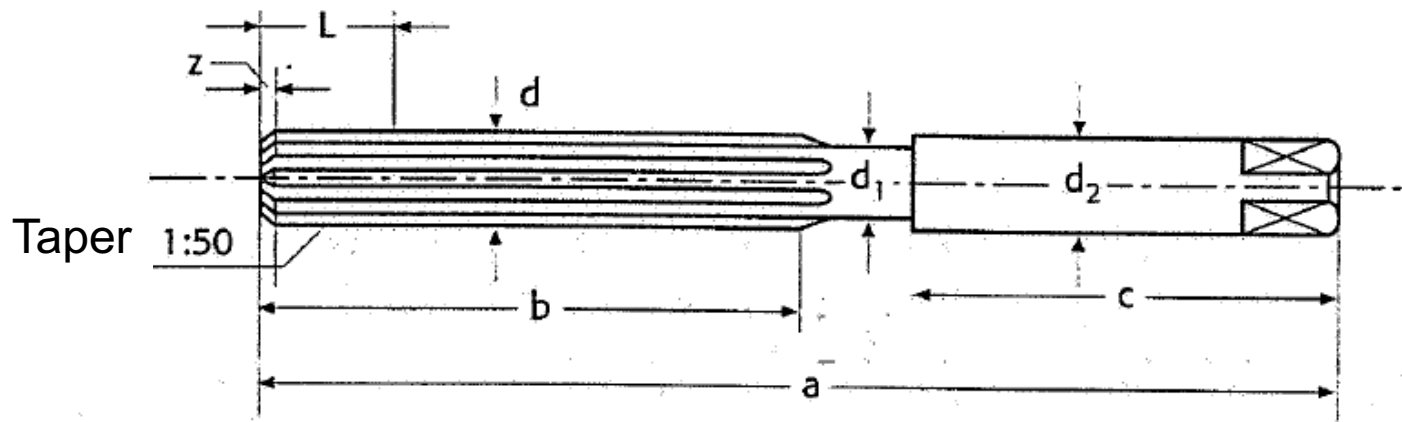


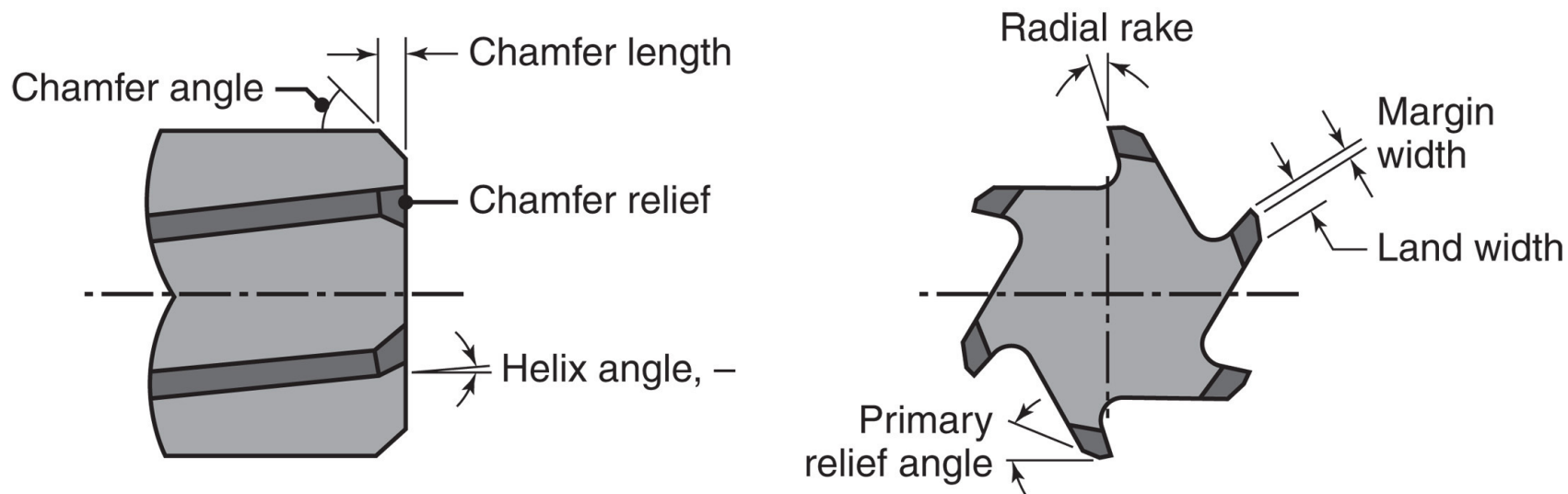
(f)



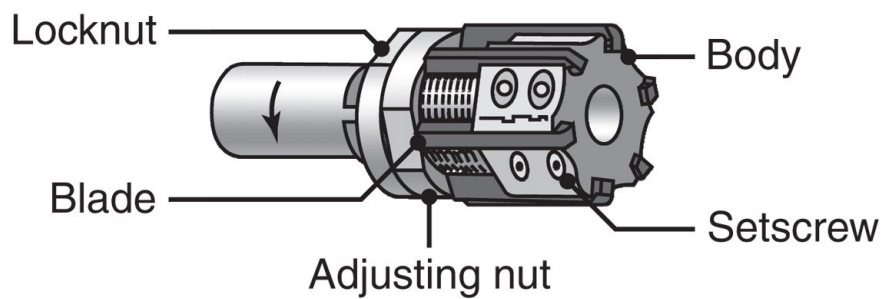
When assigned tolerances are very tight, we can use particular tool: a reamer. The operation is called reaming.

The reaming tools do not drill new holes, they only enlarge existing holes (they are slightly tapered).

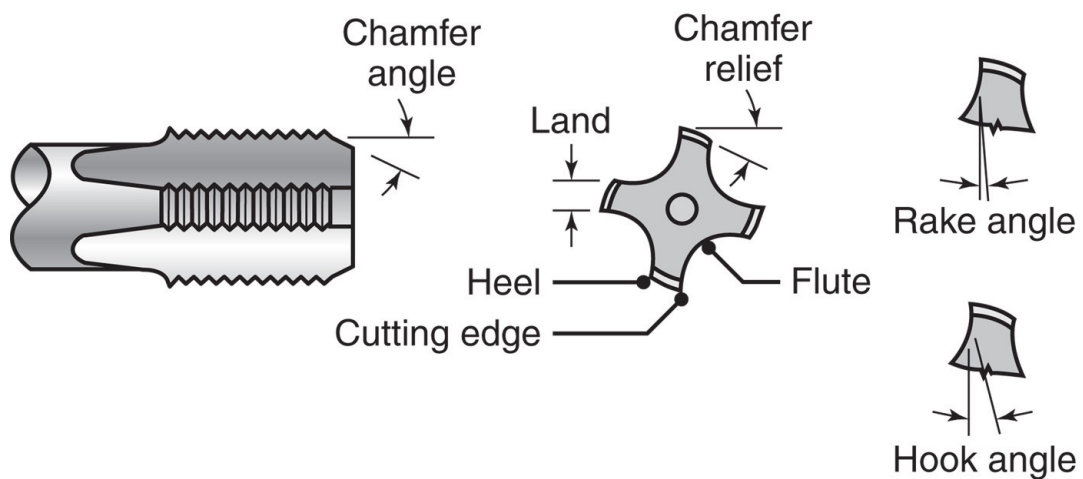




(a)



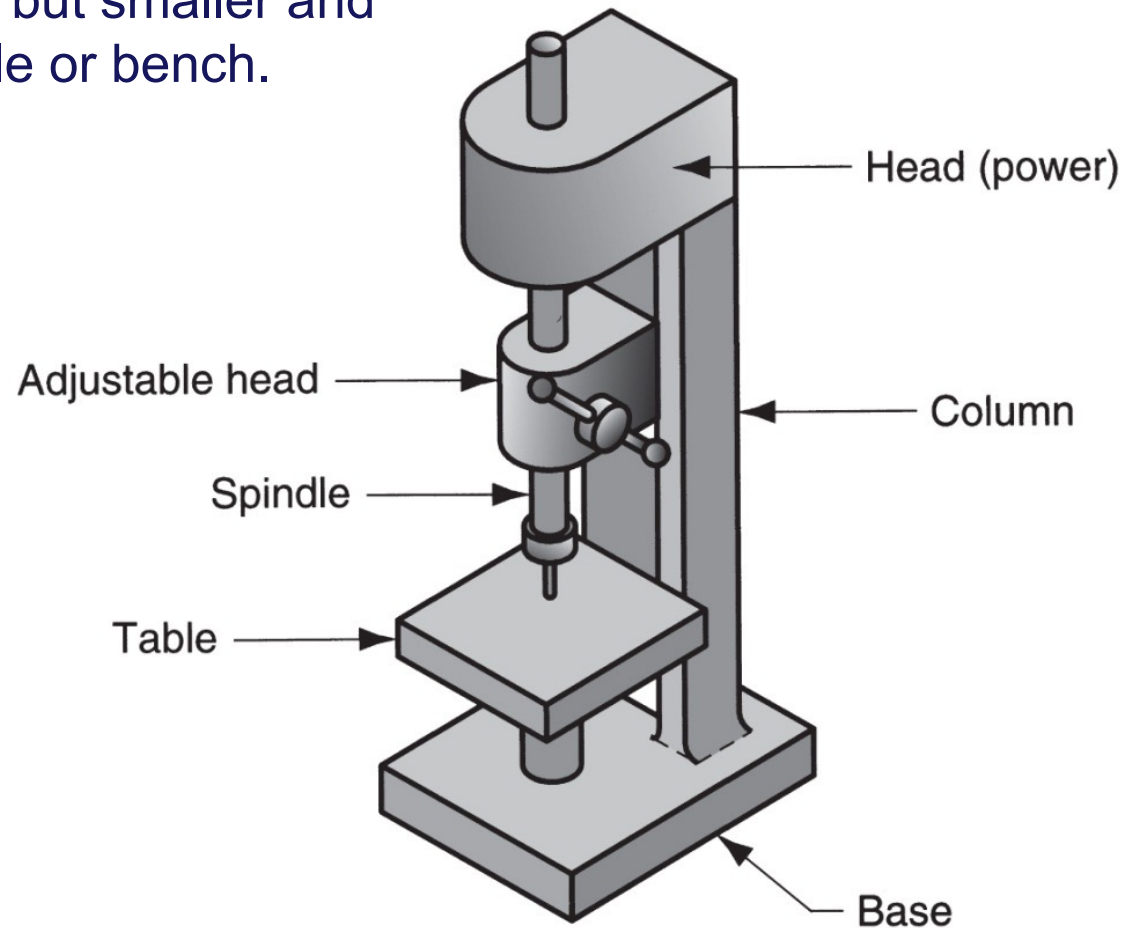
(b)





MACHINE TOOL: DRILL PRESS

Upright drill press stands on the floor.
Bench drill similar but smaller and
mounted on a table or bench.

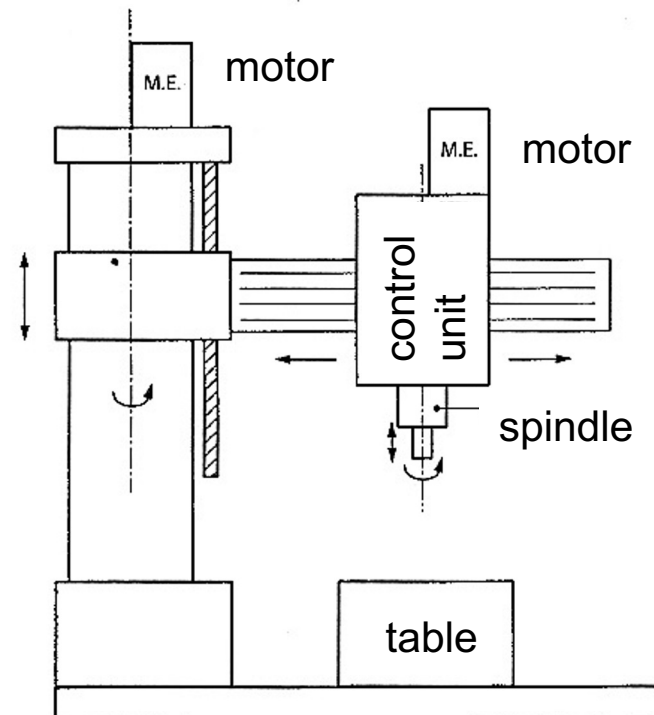
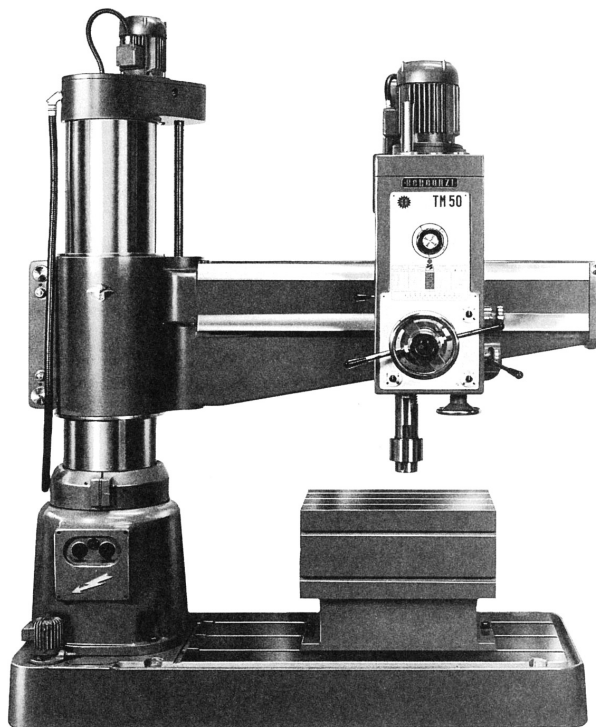




MACHINE TOOL: RADIAL DRILL PRESS

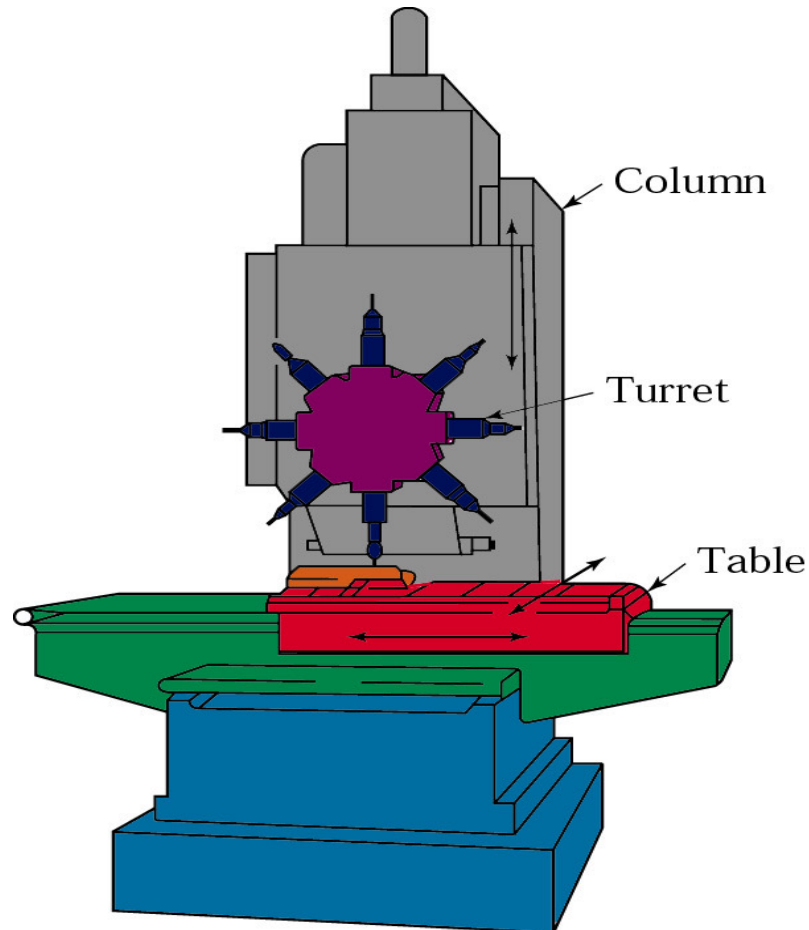
25

Large drill press designed for large parts





Three axis CNC drilling machine





WORK HOLDING FOR DRILL PRESSES

Workpiece in drilling can be clamped in any of the following workholders:

- *Vise* - general purpose workholder with two jaws
- *Fixture* - workholding device that is usually custom-designed for the particular workpiece
- *Drill jig* – similar to fixture but also provides a means of guiding the tool during drilling

